

MT WASHINGTON REGIONAL, NH, USA

WMO: 726164

Lat: 44.368N

Lon: 71.545W

Elev: 1074

StdP: 14.13

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 54728

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.5	-10.7	-23.7	1.5	-15.7	-17.5	2.2	-9.3	20.1	26.1	18.2	21.7	0.6	210	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.8	85.9	70.1	82.6	68.1	80.7	66.8	72.4	81.8	70.7	79.5	69.0	77.0	6.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.8	114.2	77.0	67.7	106.1	75.3	65.9	99.4	73.9	36.7	81.9	35.2	79.1	33.7	77.2	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	15.5	12.7	-25.9	90.4	7.3	2.5	-31.2	92.1	-35.5	93.6	-39.7	94.9	-45.0	96.7
			-25.4	75.4	6.3	1.7	-30.0	76.6	-33.7	77.6	-37.3	78.5	-41.9	79.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.0	15.0	17.4	27.1	40.9	52.8	61.0	65.9	63.9	56.9	45.3	34.2	22.1
	DBStd	20.00	13.35	12.30	11.30	8.38	7.68	6.60	5.45	5.69	7.48	8.27	9.43	11.39
	HDD50	4621	1086	914	712	296	56	2	0	0	19	195	477	864
	HDD65	8592	1551	1334	1175	723	385	156	56	90	259	611	923	1329
	CDD50	1705	0	0	2	23	144	332	492	431	227	50	4	0
	CDD65	198	0	0	0	0	7	35	83	56	17	0	0	0
	CDH74	2754	0	0	2	31	195	537	1016	694	268	11	0	0
	CDH80	642	0	0	0	4	38	142	253	145	60	0	0	0
	WSAvg	3.9	3.9	4.5	5.0	5.3	4.5	3.7	3.1	2.7	2.8	3.9	3.8	3.8
	PrecAvg	41.9	2.6	2.2	2.5	3.1	3.8	4.4	4.3	4.5	3.5	3.9	3.3	3.3
(15) Precipitation	PrecMax	52.6	4.8	4.7	3.9	6.6	6.0	8.5	7.7	8.1	7.8	9.8	5.8	5.8
	PrecMin	30.1	0.7	0.7	1.1	0.9	1.3	1.3	2.0	1.7	1.3	0.7	0.7	1.5
	PrecStd	5.3	0.9	0.9	0.8	1.4	1.4	1.9	1.3	1.8	1.3	2.3	1.1	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.4	54.0	64.2	78.7	84.4	88.7	89.5	88.1	86.6	75.2	65.5	54.2
		MCWB	49.3	46.3	53.4	58.9	65.8	70.7	73.4	71.6	70.2	62.6	56.3	50.0
	2%	DB	42.5	45.3	54.9	71.6	79.4	83.9	85.6	83.6	80.9	70.5	60.4	46.8
		MCWB	38.4	38.8	44.8	55.2	61.1	68.7	70.7	69.4	67.6	59.8	52.7	43.6
	5%	DB	36.9	39.1	48.3	64.2	74.8	80.7	82.4	81.1	76.6	65.6	54.4	41.7
		MCWB	34.1	34.6	40.7	50.4	59.3	66.1	68.6	68.0	65.4	56.9	48.4	38.9
	10%	DB	33.9	35.5	43.3	57.5	70.3	76.0	80.4	78.5	72.5	60.9	48.9	37.2
		MCWB	31.8	32.1	37.1	46.9	57.2	63.4	67.3	66.5	63.0	54.2	44.3	34.5
	0.4%	WB	49.5	48.4	53.7	62.2	68.2	73.3	75.5	73.9	71.7	65.3	58.7	50.5
		MCDB	52.2	53.6	62.8	73.4	80.1	83.8	85.1	83.0	81.9	70.9	64.0	53.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	38.7	39.6	46.6	57.3	65.0	71.0	72.9	71.5	69.2	61.4	54.1	44.1
		MCDB	41.2	43.8	52.2	68.7	75.0	80.8	81.9	80.6	78.0	67.9	59.2	46.6
	5%	WB	34.6	35.2	41.4	52.6	62.0	68.5	70.9	69.8	67.1	58.2	49.5	38.9
		MCDB	36.5	38.3	47.4	61.7	71.1	76.8	79.4	77.3	74.5	64.5	53.9	41.4
	10%	WB	31.6	32.5	37.6	48.2	59.3	65.8	69.0	68.1	64.5	54.7	44.9	35.1
		MCDB	33.9	35.2	42.5	57.2	67.5	73.0	76.7	75.5	70.4	60.0	48.4	37.0
(35) Mean Daily Temperature Range	5% DB	MDBR	21.4	24.0	22.5	23.2	25.6	24.7	24.8	24.9	25.9	21.6	18.6	18.1
		MCDBR	21.4	25.4	29.9	37.0	35.3	31.2	28.5	28.3	30.8	30.7	27.4	21.2
		MCWBR	19.0	20.3	20.4	21.5	18.9	17.0	15.0	15.6	18.5	20.6	20.5	18.9
	5% WB	MCDBR	21.2	23.1	25.9	31.6	28.0	25.6	24.4	24.0	26.3	26.6	24.7	20.8
		MCWBR	19.1	18.9	19.0	20.3	16.0	14.9	13.4	13.9	16.4	19.2	20.0	19.0
(40) Clear-Sky Solar Irradiance	taub		0.256	0.276	0.310	0.344	0.366	0.412	0.420	0.394	0.358	0.322	0.303	0.269
	taud		2.421	2.336	2.358	2.403	2.402	2.266	2.254	2.321	2.416	2.526	2.514	2.459
	Ebn at Noon		281	292	294	292	287	272	268	272	275	273	260	263
	Edh at Noon		23	30	34	35	36	42	42	38	32	25	21	20
(44) All-Sky Solar Radiation	RadAvg		467	741	1109	1372	1558	1699	1728	1568	1277	768	485	357
	RadStd		52	81	72	164	189	149	141	90	89	81	60	30

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			+1.19	N/A	N/A	N/A	N/A	N/A	-441	+254	+58
(47) Regional (51 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	+126	N/A

Nomenclature: See separate page